Curated Intermediate Worksheet: Integers

Focus Area: BODMAS Simplification (Exercise 1D) & Word Problems (Exercise 1E)

Core Rule Recap: BODMAS & Sign Laws

BODMAS Rule Order of Operations:

1. **B (Brackets):** In sequence of: (i) Bar or Vinculum ($\overline{a-b}$), (ii) Parentheses ( ), (iii) Braces { }, (iv) Square Brackets [ ].
2. **O (Of):** Equivalent to multiplication, solved immediately after brackets.
3. **D / M / A / S:** Division ($\div$), Multiplication ($\times$), Addition ($+$), Subtraction ($-$).

Word Problem Conventions: East ($+$) / West ($-$), Depth/Sea-level (0), Descent ($-$) / Ascent ($+$), Profit ($+$) / Loss ($-$).

Section A: Multiple Choice Questions (10 Marks)

Choose the correct option. Each question carries 1 mark.

1. Simplify the expression: $(-18) \div [(-5) + 3] - (-4)$.

(a) 13
(b) 5
(c) -13
(d) -5

2. Evaluate the expression: $12 - [8 - \{15 - (6 - 4)\}]$.

(a) 1
(b) 17
(c) 7
(d) -3

3. Solve using order of operations: $15 - \{12 \div (5 - \overline{3 - 1})\}$.

(a) 11
(b) 13
(c) 14
(d) 9

4. A submarine is at a depth of 450 m below sea level. If it ascends 120 m and then dives 230 m, what is its new depth with respect to sea level?

(a) 560 m below sea level
(b) 100 m below sea level
(c) 340 m below sea level
(d) 780 m below sea level

5. On a particular day, the morning temperature of Leh was $-14°C$. By afternoon, it rose by $8°C$ and by midnight, it fell by $11°C$. What was the temperature at midnight?

(a) $-3°C$
(b) $-17°C$
(c) $-11°C$
(d) $-25°C$

6. An elevator descends into a mine shaft at a constant rate of 6 m/min. If the descent starts from 20 m above the ground level, how long will it take to reach -220 m?

(a) 40 minutes
(b) 30 minutes
(c) 45 minutes
(d) 36 minutes

7. A shopkeeper earns a profit of Rs. 5 on every notebook sold and incurs a loss of Rs. 2 on every pen sold. In a month, she sold 40 notebooks and some pens, making a net profit of Rs. 120. How many pens did she sell?

(a) 30
(b) 40
(c) 50
(d) 60

8. Simplify the nested expression: $(-28) \div \{10 - [(-12) \div (-4)]\}$.

(a) -4
(b) 4
(c) -7
(d) 7

HOTS QUESTION (High Order Thinking Skills)

Let the symbol $\otimes$ define a binary operation on integers such that $a \otimes b = a \times b - (a + b)$. Find the value of $(-5) \otimes [(-2) \otimes 3]$.

9. (a) 47

(b) 23

(c) -47

(d) -23

HOTS QUESTION (High Order Thinking Skills)

A digital thermometer displays temperature in Celsius (°C). If the temperature T satisfies the relation $T = -12 \div [x - \{15 \div (8 - \overline{5 - 2})\}]$ and the display shows -4°C, find the value of x.

10. (a) 6

(b) 0

(c) -3

(d) 9

Section B: Subjective Problems (30 Marks)

Solve the following problems with step-by-step mathematical reasoning.

1. (3 Marks) Simplify the expression:
- [18 - {16 - (12 $\div \overline{5 - 1})\}$

]

2. (3 Marks) Evaluate the expression using the BODMAS rule:
(-15) - [(-6) - {24 $\div (10 - \overline{8 - 6})\}$

]

3. (3 Marks) A freezer's temperature starts at 22°C and drops at a constant rate of 4°C every 15 minutes. How many hours will it take for the temperature to reach -10°C?

4. (3 Marks) Simplify the expression:
[(-100) $\div (-5)$
$\div [(-8) \div 2] - \{(-6) \times (-3)\}$
]

5. (3 Marks) A shopkeeper earns a profit of Rs. 8 on selling each large chocolate box and incurs a loss of Rs. 3 on selling each small candy packet. On Monday, he sold 15 large chocolate boxes. If his net profit for that day was Rs. 45, find the number of small candy packets he sold.

6. (3 Marks) A hiker starts at an altitude of 1850 m above sea level and climbs down 120 m into a valley. From there, she climbs up 340 m to a ridge, and then goes down 520 m to reach her campsite. What is the altitude of her campsite with respect to sea level?

7. (3 Marks) Simplify:
- [38 - {60 $\div (3 \times \overline{9 - 5})\}$

]

8. **(3 Marks)** In a class test of 20 questions, 5 marks are awarded for every correct answer, -2 marks for every incorrect answer, and 0 marks for unattempted questions.

(a) Kavita scored 59 marks and got 13 correct answers. How many questions did she attempt incorrectly?

(b) Rahul attempted 11 questions and scored a net score of -1 mark. How many correct answers did he get?

HOTS QUESTION (High Order Thinking Skills)

(3 Marks) A deep-sea research probe is programmed to dive and ascend according to a sequential algebraic program. At time $t = 0$, its depth is -150 m (with respect to sea level). It then executes a repeating cycle of operations: it descends 45 m in the first minute, descends 30 m in the second minute, and ascends 15 m in the third minute.

9. (a) What is the probe's depth after 1 cycle (3 minutes)?

(b) How many complete cycles must it run to end at a depth of -390 m? Does it touch or exceed this depth at any point before the end of the final cycle?

HOTS QUESTION (High Order Thinking Skills)

(3 Marks) Let the mathematical operation $\mathcal{H}(a, b, c)$ be defined for any integers a, b, c as:
H(a, b, c) = (a $\times b) - \{(b \times c) \div \overline{a - c}\}$

10. (a) Evaluate $\mathcal{H}(-4, -6, -2)$.

(b) If $\mathcal{H}(2, y, -2) = -15$, find the integer value of y.

Detailed Solutions Key

Section A: Multiple Choice Solutions

1. **Correct Option: (a)**
   Expression: $(-18) \div [(-5) + 3] - (-4)$.
   Simplify brackets: $(-5) + 3 = -2$.
   Then, division: $(-18) \div (-2) = 9$.
   Finally, subtraction: $9 - (-4) = 9 + 4 = 13$.

2. **Correct Option: (b)**
   Expression: $12 - [8 - \{15 - (6 - 4)\}]$.
   Parentheses first: $6 - 4 = 2$.
   Then curly braces: $15 - 2 = 13$.
   Then square brackets: $8 - 13 = -5$.
   Finally: $12 - (-5) = 12 + 5 = 17$.

3. **Correct Option: (a)**
   Expression: $15 - \{12 \div (5 - \overline{3 - 1})\}$.
   Evaluate bar/vinculum first: $\overline{3 - 1} = 2$.
   Then parentheses: $5 - 2 = 3$.
   Then curly braces (division): $12 \div 3 = 4$.
   Finally: $15 - 4 = 11$.

4. **Correct Option: (a)**
   Represent depth as a negative value: -450 m.
   Ascending is positive ($+120$ m); diving is negative (-230 m).
   Calculation: $-450 + 120 - 230 = -330 - 230 = -560$ m.
   This represents a depth of 560 m below sea level.

5. **Correct Option: (b)**
   Morning temperature: $-14°$C.
   Rising temperature is added ($+8°$C); falling temperature is subtracted ($-11°$C).
   Midnight temperature: $-14 + 8 - 11 = -6 - 11 = -17°$C.

6. **Correct Option: (a)**
   Start altitude: $+20$ m. Target altitude: -220 m.
   Total distance to descend: $20 - (-220) = 20 + 220 = 240$ m.
   Descent rate: 6 m/min.
   Time required: $240 \div 6 = 40$ minutes.

7. **Correct Option: (b)**
   Let the number of pens sold be p.
   Profit from notebooks: $40 \times \text{Rs. } 5 = \text{Rs. } 200$.
   Loss from pens: $p \times \text{Rs. } 2 = 2p$.
   Net profit equation: $200 - 2p = 120 \implies 2p = 80 \implies p = 40$.
   Thus, she sold 40 pens.

8. **Correct Option: (a)**
   Expression: $(-28) \div \{10 - [(-12) \div (-4)]\}$.
   Square brackets first: $(-12) \div (-4) = 3$.
   Curly braces next: $10 - 3 = 7$.
   Final division: $(-28) \div 7 = -4$.

9. **Correct Option: (a)**
Binary operation: $a \otimes b = a \times b - (a + b)$.
First evaluate $(-2) \otimes 3$:
$(-2) \otimes 3 = (-2 \times 3) - (-2 + 3) = -6 - 1 = -7$.
Now evaluate $(-5) \otimes (-7)$:
$(-5) \otimes (-7) = (-5 \times -7) - (-5 + -7) = 35 - (-12) = 35 + 12 = 47$.

10. **Correct Option: (a)**
Equation: $-4 = -12 \div [x - \{15 \div (8 - \overline{5 - 2})\}]$.
Solve bar: $\overline{5 - 2} = 3$.
Solve braces: $15 \div (8 - 3) = 15 \div 5 = 3$.
Substitute back: $-4 = -12 \div [x - 3] \implies [x - 3] = -12 \div (-4) \implies x - 3 = 3 \implies x = 6$.

Section B: Subjective Solutions

1. **Step-by-step simplification:**
Expression: $32 - [18 - \{16 - (12 \div \overline{5 - 1})\}]$.
1. Resolve the bar/vinculum: $\overline{5 - 1} = 4$.
2. Solve round brackets: $12 \div 4 = 3$.
3. Solve curly braces: $16 - 3 = 13$.
4. Solve square brackets: $18 - 13 = 5$.
5. Final subtraction: $32 - 5 = 27$.
Final Answer: 27

2. **Step-by-step simplification:**
Expression: $(-15) - [(-6) - \{24 \div (10 - \overline{8 - 6})\}]$.
1. Resolve the bar: $\overline{8 - 6} = 2$.
2. Solve inside curly braces:
Parentheses: $10 - 2 = 8$.
Division: $24 \div 8 = 3$.
3. Solve square brackets: $-6 - 3 = -9$.
4. Final expression: $-15 - (-9) = -15 + 9 = -6$.
Final Answer: -6

3. **Temperature descent modeling:**
Initial temperature $= 22°\text{C}$. Target temperature $= -10°\text{C}$.
Total required temperature change:
$\Delta T = \text{Target} - \text{Initial} = -10°\text{C} - 22°\text{C} = -32°\text{C}$.
The temperature falls by $4°\text{C}$ every 15 minutes.
Number of 15-minute intervals required $= 32 \div 4 = 8$ intervals.
Total duration $= 8 \times 15$ minutes $= 120$ minutes.
Since 60 minutes $= 1$ hour, total time $= 120 \div 60 = 2$ hours.
Final Answer: 2 hours

4. **Step-by-step simplification:**
Expression: $[(-100) \div (-5)] \div [(-8) \div 2] - \{(-6) \times (-3)\}$.
1. Evaluate first square bracket: $(-100) \div (-5) = 20$.
2. Evaluate second square bracket: $(-8) \div 2 = -4$.
3. Perform division between brackets: $20 \div (-4) = -5$.
4. Evaluate multiplication in curly braces: $(-6) \times (-3) = 18$.
5. Combine terms: $-5 - 18 = -23$.
Final Answer: -23

5. **Profit and Loss equation:**
Let the number of small candy packets sold be x.
Profit from 15 large chocolate boxes $= 15 \times \text{Rs. } 8 = \text{Rs. } 120$.
Loss from x small candy packets $= x \times \text{Rs. } 3 = 3x$.
Net Profit $=$ Profit $-$ Loss.
$45 = 120 - 3x \implies 3x = 120 - 45 \implies 3x = 75 \implies x = 25$.
Final Answer: 25 small candy packets

6. **Altitude vector tracking:**
Starting altitude $= +1850$ m.
1. Descend 120 m: $1850 - 120 = 1730$ m.
2. Climb up 340 m: $1730 + 340 = 2070$ m.
3. Descend 520 m: $2070 - 520 = 1550$ m.
The campsite altitude is 1550 m above sea level.
Final Answer: 1550 m above sea level

7. Step-by-step simplification:
Expression: $45 - [38 - \{60 \div (3 \times \overline{9 - 5})\}]$.
1. Resolve the bar: $\overline{9 - 5} = 4$.
2. Evaluate parentheses: $3 \times 4 = 12$.
3. Evaluate curly braces: $60 \div 12 = 5$.
4. Evaluate square brackets: $38 - 5 = 33$.
5. Final subtraction: $45 - 33 = 12$.

Final Answer: 12

8. Multi-part grading scheme analysis:
Correct answer = $+5$, Incorrect answer = -2, Unattempted = 0.

(a) Kavita scored 59 marks with 13 correct answers.
Let the number of incorrect answers be w.
Total marks $= (13 \times 5) + w \times (-2) = 59$.
$65 - 2w = 59 \implies 2w = 6 \implies w = 3$.
She attempted **3 questions incorrectly**.

(b) Rahul attempted 11 questions and scored -1 mark.
Let c be correct answers; then incorrect answers are $(11 - c)$.
Total marks $= 5c - 2(11 - c) = -1$.
$5c - 22 + 2c = -1 \implies 7c = 21 \implies c = 3$.
He got **3 correct answers**.

9. Deep-sea probe cyclic motion:

(a) Initial depth $= -150$ m.
1 cycle consists of 3 minutes:
Minute 1: Descend 45 m ($\Delta y = -45$).
Minute 2: Descend 30 m ($\Delta y = -30$).
Minute 3: Ascend 15 m ($\Delta y = +15$).
Net change per cycle $= -45 - 30 + 15 = -60$ m.
Depth after 1 cycle $= -150 - 60 = -210$ m (or 210 m depth).

(b) Target final depth $= -390$ m.
Each complete cycle changes depth by -60 m.
To go from -150 m to -390 m, total net change required $= -390 - (-150) = -240$ m.
Number of complete cycles required $= -240 \div (-60) = 4$ cycles.
Let us check if the probe touches or exceeds -390 m depth during the 4th cycle:
At the start of the 4th cycle (after 3 cycles / 9 minutes), depth is:
$-150 + 3 \times (-60) = -330$ m.
During the 4th cycle:
- Minute 10 (1st minute of cycle 4): Probe descends 45 m $\implies$ depth becomes $-330 - 45 = -375$ m.
- Minute 11 (2nd minute of cycle 4): Probe descends 30 m $\implies$ depth becomes $-375 - 30 = -405$ m.
Since -405 m is deeper than -390 m, the probe **does touch and exceed** the depth of -390 m during the 11th minute (2nd minute of the 4th cycle), before ascending back to -390 m at the end of the 12th minute.

10. Mathematical operation evaluation:
Formula: $\mathcal{H}(a, b, c) = (a \times b) - \{(b \times c) \div \overline{a - c}\}$.

(a) Evaluate $\mathcal{H}(-4, -6, -2)$:
Identify terms: $a = -4, b = -6, c = -2$.
- $\overline{a - c} = \overline{-4 - (-2)} = -4 + 2 = -2$.
- $b \times c = -6 \times (-2) = 12$.
- $\{(b \times c) \div \overline{a - c}\} = 12 \div (-2) = -6$.
- $a \times b = -4 \times (-6) = 24$.
- $\mathcal{H}(-4, -6, -2) = 24 - \{-6\} = 24 + 6 = 30$.

(b) Solve for y in $\mathcal{H}(2, y, -2) = -15$:
Identify terms: $a = 2, c = -2$.
- $\overline{a - c} = \overline{2 - (-2)} = 4$.
- $b \times c = y \times (-2) = -2y$.
- $\{(b \times c) \div \overline{a - c}\} = -2y \div 4 = -\frac{y}{2}$.
- $a \times b = 2y$.
- $\mathcal{H}(2, y, -2) = 2y - \left(-\frac{y}{2}\right) = 2y + \frac{y}{2} = \frac{5y}{2}$.
We are given: $\frac{5y}{2} = -15 \implies 5y = -30 \implies y = -6$.